Assignment-1

Note: All Question must be compulsory and Write questions with take full Screen short. Make PDF. In Every Page write roll number, name and division.

1. List up to IO different protocols that appear in the protocol column in the unfiltered packet-listing window.
2. What is the Internet address of the www.gujaratuniversity.ac.in? What is the Internet address of your computer?
3. http protocol only with Wireshark running on your computer.
4. How many packets did you capture (total of all protocols, not just HTTP)?
5. Now, use display filters to determine how many packets contain your ip address (hint: Use ip.addr instead of the clumsy ip.src or ip.dst format).
6. What is this filter you used? Now, reverse the filter to determine how many packets don't contain your ip address.
7. What is the IP address of your computer? Of the **gtu.ac.in** server?
8. What HTTP version is your browser running? What version of HTTP is the server running?(HTTP browser **request** line (GET / HTTP/1.1)& HTTP server **response** line (HTTP/1.1 200 OK)(http.request.method == GET)
9. What is the status code and phrase returned from the server to your browser?(http.response.code)
10. What languages does your browser indicate to the server that it can accept? Which header line is used to indicate this information?
    http.request or http.accept_language
11. When was the HTML-file, that you have retrieved, last modified at the server? Which header line is used to indicate this information?
    http.response.code == 200 && http.last_modified
12. How many bytes of content (size of file) are returned to your browser? Which header line is used to indicate this information?

HTTP Conditional GET/Response Interaction

13. Start up the Wireshark packet sniffer, and make sure that "http" is in the displayfilter, so that only captured HTTP messages will be displayed in the packet-list pane.
14. Enter the following URL into your browser:
    http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html

Your browser should display a very simple HTML file.

15. Quickly enter the same URL into your browser again (or simply select the refresh button on your browser).
16. Stop Wireshark packet capture.

Answer the following questions. Please explain how you can find the answers

17. Inspect the contents of the first HTTP GET request from your browser to the server. Is there an "IF-MODIFIED-SINCE" header line in the HTTP GET message? Why or why not?
18. Inspect the contents of the server response. Has the server explicitly returned the contents of the file? How can you tell?
19. Now inspect the contents of the second HTTP GET request from your browser to the server. Is there an "IF-MODIFIED-SINCE:" header line in the HTTP GET message? If so, what information follows the "IF-MODIFIED-SINCE:" header line?
20. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Has the server explicitly returned the contents of the file? Explain.

21. How many HTTP GET request messages has your browser sent? Which packet in the trace contains the request for the Bill of Rights?
    Note: The packets here are only those containing HTTP messages and data.
22. Which packet in the trace contains the status code and phrase associated with the response to the HTTP GET request? What is the status code and phrase in the response?
23. Discuss http GET and RESPONSE
24. Count the total number of HTTP GET requests.
    Ans.: Enter in filter: http.request.method==GET , then count total ------ marked.
25. How much time elapsed between the first HTTP GET request from client and the HTTP response (OK) from server?
    Ans.: http.request.method==GET || http.response
26. Identify the source and destination ports used to transfer the first HTTP GET request
27. Identify the client and server ip address in the first HTTP GET request?
    Ans.: Apply filter http.request.method==GET, see the IP , within this you will find source ip address and destination ip address.
    28. Identify the client and server MAC address in the first HTTP GET request?
    Ans.: Apply filter http.request.method==GET, see the Ethernet , within this you will find source MAC and destination MAC.

DNS PROTOCOL

Start Wireshark packet capture.
- Open your browser and visit https://www.amazon.com/
This generates DNS packets because the browser first requests the IP address of `amazon` from the DNS server.

- After the website opens successfully, **stop the Wireshark packet capture.**

1. Use the filtering functionality of Wireshark by typing "dns" in the filtering field, new window will appear:
2. Let's try now to find out what are those packets contain by following one of the conversations (also called network flows), select one of the packets and press the right mouse button, you should see another screen appear.
   Click on Follow UDP Stream, and then you will see another screen showing content https://www.amazon.com/
3. At Command prompt ; Nslookup https://www.amazon.com/
4. Locate the DNS Query and Response message, Are they sent on UDP or TCP
5. What is the destination port for the DNS query message ? What is the source port of of DNS response message?

6. What is the source IP address of the DNS Query message? Is it the IP address of your computer? Verify using `ipconfig`. Are these two IP address the same?

7. What is the destination IP address of the DNS Query message? Is it the IP address of your local DNS server? Verify using `ipconfig/all`. Are these two IP address the same?

8. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?
9. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?
   The response DNS message contains one answer containing information about the name of the host, the type of address, class, the TTL,the data length .
10. What Protocol use in Payload? How many bytes use in Payload?
11. Find **stream index** and **stream packet number** of DNS Response.

TCP Protocol:

- Start up your web browser. Go the http://gaia.cs.umass.edu/wireshark-labs/alice.txt and retrieve an ASCII copy of *Alice in Wonderland*. Store this file somewhere on your computer.
- Next go to http://gaia.cs.umass.edu/wireshark-labs/TCP-wireshark-file1.html.

Answer the question:

1. What is the IP address and TCP port number used by the client computer (source) that is transferring the file to gaia.cs.umass.edu?

To answer this question, the easiest way is to select the HTTP POST message (or any HTTP message of the same file upload) in Wireshark. Then, open the TCP packet details. In the packet details window, you can find the client's IP address and TCP port number.

2.What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?

3. **What is the IP address and TCP port number used by your client computer (source) to transfer the file to gaia.cs.umass.edu?**

4.What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? What is it in the segment that identifies the segment as a SYN segment?

5. What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value? What is it in the segment that identifies the segment as a SYNACK segment?

6. **Find the TCP packet that contains the HTTP POST command. What is its Sequence Number?**

To find the HTTP POST command, look at the packet contents in the bottom pane of the Wireshark window and find the packet whose Data field contains the word "POST".

7. What is the length of each of the first six TCP segments?

https://www.myntra.com/

UDP Protocol:

1. Select *one* UDP packet from your trace. From this packet, determine how many fields there are in the UDP header.Name these fields.
   Determine the length (in bytes) of the each of the udp header fields.

2. The value in the Length field is the length of what? (You can consult the text for this answer). Verify your claim with your captured UDP packet.

3. What is the maximum number of bytes that can be included in a UDP payload? (Hint: the answer to this question can be determined by your answer to 2. above)

4.What is the largest possible source port number? (Hint: see the hint in 3.)

5. What is the protocol number for UDP? Give your answer in both hexadecimal and decimal notation. To answer this question, you'll need to look into the Protocol field of the IP datagram containing this UDP segment .

6. Examine a pair of UDP packets. The first packet is sent by your host, and the second packet is the reply to the first packet. Compare the source and destination port numbers of both packets. What is the relationship between them?

Icmp protocol

- Open Command Prompt (CMD) and Wireshark. Start packet capture in Wireshark.
- Run the command `ping -n 10 www.sandesh.com`.
- After all 10 ping requests and replies are completed, stop the capture.
- Apply the `icmp` filter and use the same captured packets to answer all the ICMP (Ping) practical questions.

1.What is the IP address of your host? What is the IP address of the destination host?

2.Why doesn't an ICMP packet have source and destination port numbers?

3. Examine one of the ping request packets sent by your host. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

4. What is the IP address of your host? What is the IP address of the target destination host? (using tracert command)

5) If ICMP uses UDP packets instead, will the Protocol Number still be 1? If not, what will it be?

6)Look at the last three ICMP packets received by your computer. How are they different from the ICMP Error packets? Why are they different?

7) What is the ICMP Type and Code for Echo Reply

8) What is the purpose of the `tracert` command? What is the purpose of the `ping` command?